GPT NLP Lab Assistant Group 3

Michael Fritzer, Muhammed Faseeh,
David Haybarger, Kevin Lee

INTERFACE CONTROL DOCUMENT

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Interface Control Document
for
GPT NLP Lab Assistant Group 3

Team AURA

David Haybarger 09/30/2025

Author Date

Approved by:

Michael Fritzer 12/06/2025

Project Leader Date

Byung-Jun Yoon 12/06/2025

Professor Date

Fahrettin Ay 12/06/2025

T/A Date

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1. Overview

The Interface Control Document (ICD) for the Autonomous University Robot Assistant (AURA) defines the interfaces between its hardware, software, and peripheral subsystems. It specifies the physical, thermal, electrical, and communication requirements required for proper integration and operation of the robot. This document establishes compatibility across all subsystems and serves as a reference for assembly, maintenance, troubleshooting, and future scaling of the platform.

2. References and Definitions

2.1. References

Refer to section 2.2 of the Functional System Requirements document.

2.2. Definitions

AI	Artificial Intelligence
AURA	Autonomous University Robot Assistant
BIT	Built In Test
CCA	Circuit Card Assembly
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ESP32	Microcontroller for wireless connectivity and low level control
FOV	Field of View
GPT	Generative Pretrained Transformer
GPIO	General Purpose Input/Output
GUI	Graphical User Interface
Hz	Hertz
ICD	Interface Control Document
Jetson Orin Nano	Embedded AI computer responsible for system processing
kHz	Kilohertz (1,000 Hz)
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LLM	Large Language Model
mA	Milliamp
MCU	Motor Control Unit
MHz	Megahertz (1,000,000 Hz)
mW	Milliwatt
NLP	Natural Language Processing
PCB	Printed Circuit Board
PSU	Power Supply Unit
PWM	Pulse Width Modulation
RAG	Retrieval Augmented Generation
SAM	Segment Anything Model
SOP	Standard Operating Procedure
STT	Speech to Text
TBD	To Be Determined
TA	Teaching Assistant
TTS	Text to Speech
UART	Universal Asynchronous Receiver/Transmitter
UI	User Interface
USB	Universal Serial Bus

3. Physical Interface

3.1. Weight

3.1.1. Main Processing Unit and Power System

Component	Weight	Number of Items	Total Weight
Jetson Orin Nano	222.26g	1	222.26g
Battery Power Supply	152.12g	1	152.12g
Custom Power PCB	83.22g	1	83.22g

Table 1: Main Processing Unit and Power System Weight

3.1.2. Robotic System

Component	Weight	Number of Items	Total Weight
Servo Motors	62g	3	186g
ESP32 MCU	6g	1	6g
Servo Microcontroller	9g	1	9g

Table 2: Robotic System Weight

3.1.3. UI Components

Component	Weight	Number of Items	Total Weight
Digital Screen	113g	1	113g
Camera	3g	1	3g
Microphone	8g	1	8g

Table 3: UI Components Weight

3.1.4. 3D Printed Shell

Component	Weight	Number of Items	Total Weight
Body & Arms	854g	3	854g

Table 4: 3D Printed Shell Weight

3.2. Dimensions

3.2.1. Dimensions of Main Processing Unit and Power System

Component	Length	Width	Height
Jetson Orin Nano	100mm	80.5mm	48mm
Battery Power Supply	85mm	15mm	43mm
Custom Power PCB	118mm	38mm	3mm

Table 5: Main Processing Unit and Power System Dimensions

3.2.2. Dimensions of Robotic System

Component	Length	Width	Height
Servo Motors	54.5mm	20mm	47.5mm
ESP32 MCU	86mm	56mm	16mm

Table 6: Robotic System Dimensions

3.2.3. Dimensions of UI Components

Component	Length	Width	Height
Digital Screen	76.2mm	119.38mm	25.4mm
Camera	24mm	25mm	4mm
Microphone	6mm	6mm	3mm

Table 7: UI Components Dimensions

3.2.4. Dimensions of 3D Printed Shell

Component	Length	Width	Height
Body & Arms	10”	9.75”	2.75”

Table 8: 3D Printed Shell Dimensions

3.3. Mounting Locations

The Autonomous University Robot Assistant (ARUA) will be primarily deployed in university laboratory environments such as electrical engineering teaching labs. The system will be positioned near or on student workbenches where it has access to both power outlets and wireless connectivity for integration with the campus network.

Figure 1: Reference of ARUA Deployment Location

4. Thermal Interface

The Jetson Orin Nano will require a heat sink and optional fan to ensure reliable performance during extended computation. While the ESP32 microcontroller and motor driver circuits will also be exposed to similar lab conditions, their thermal output is minimal and does not require dedicated cooling. Because the Jetson Orin Nano is responsible for the majority of AI processing and higher power consumption, it is the primary component at risk of overheating. Proper ventilation slots will be included in the enclosure, and a small cooling fan may be added to maintain stable operating temperatures during long-duration lab sessions.

5. Electrical Interface

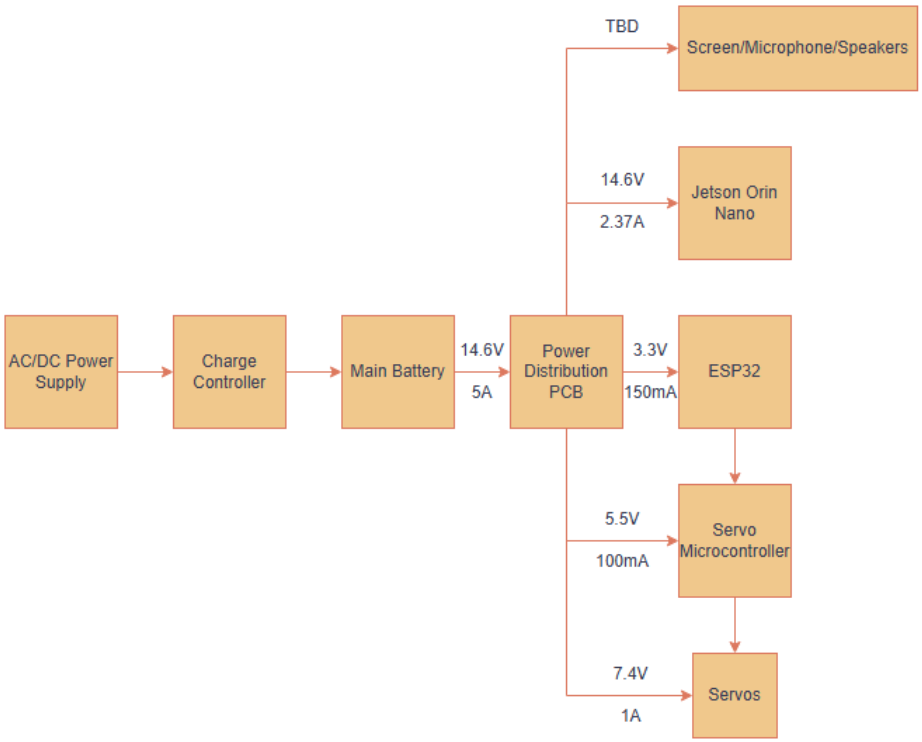

Figure 2: Electrical Interface Diagram

5.1. Primary Input Power

5.1.1 Main Power System

The Autonomous University Robot Assistant (ARUA) will be powered by a 14.6V 4-cell Li-Po battery pack rated at 5000 mAh. This battery will supply sufficient energy for several hours of continuous lab use. The power distribution system uses step-down regulators to provide 5V for servo motors, 14.6V for the Jetson Orin Nano, and 3.3V for the ESP32 and sensor interfaces.

5.1.2 Charging and Protection

An inline fuse and polarity protection circuitry are implemented to prevent damage from accidental miswiring. Battery management modules will regulate charging to ensure safe operation during repeated lab use.

5.2. User Control Interface

The user will interact with ARUA through a touchscreen interface powered by the Jetson Orin Nano. A physical emergency stop button is integrated into the chassis, providing immediate cutoff of motor power in case of malfunction. In addition, remote access is available through a web dashboard when the system is connected to Wi-Fi.

5.3. Voltage and Current Levels

5.3.1 Maximum Values

Component	Voltage (V)	Current (A)	Power (W)
Jetson Orin Nano	10.5	5	52.5
ESP32	3.3	2	6.6
Servo Motor (per unit)	7.4	0.3	2.22
Touchscreen Display	-	-	Powered via Jetson USB
Microphone	-	-	Powered via Jetson USB
Speaker	-	-	Powered via Jetson USB

Table 9: Maximum Voltage and Current Levels

Note: Touchscreen, microphone, and speaker are powered from the Jetson Orin Nano carrier board via USB and are not supplied by the custom power PCB. Their consumption is included inside the Jetson power budget.

The values in this table represent peak load conditions. These are expected only during periods of maximum computation (Jetson), simultaneous servo movement, and display usage.

5.3.2 Stand-by Values

Component	Voltage (V)	Current (A)	Power (W)
Jetson Orin Nano	7	1	7
ESP32	3.3	0.150	0.5
Touchscreen Display	-	-	Powered via Jetson USB
Microphone	-	-	Powered via Jetson USB
Speaker	-	-	Powered via Jetson USB

Table 10: Stand-by Voltage and Current Levels

Note: Touchscreen, microphone, and speaker are powered from the Jetson Orin Nano carrier board via USB and are not supplied by the custom power PCB. Their consumption is included inside the Jetson power budget.

The standby values represent average consumption under idle or low-demand operation, estimated at roughly 60–70% of total usage.

5.4 Signal Interfaces

5.4.1 PWM Motor Control

Servo motors are driven via PWM signals generated by the ESP32 and routed through motor control expansion boards. Each PWM line is isolated and regulated to prevent signal distortion.

6. Communications / Device Interface Protocols

6.1. Wireless Communications

6.1.1. Wi-Fi

The ESP32 provides IEEE 802.11 b/g/n Wi-Fi connectivity, enabling communication with the Jetson Orin Nano and external lab networks. This allows ARUA to interface with cloud services, local data servers, and remote terminals for monitoring.

6.1.2. Bluetooth

The ESP32 includes Bluetooth 4.2/5.0 support for optional low-power communication with external devices such as smartphones, tablets, or Bluetooth-enabled lab instruments.

6.2. Host Device

The Jetson Orin Nano includes multiple USB 3.0 ports used for connecting peripherals such as a keyboard, mouse, or debugging interface. These ports also support USB-to-serial adapters for direct sensor integration.

6.3. Video Interface

The Jetson Orin Nano provides HDMI 2.0 output to drive the integrated touchscreen display. An external monitor may also be connected for demonstrations or lab presentations.

6.4. Device Peripheral Interface

Communication between the ESP32 and Jetson Orin Nano is handled via UART, using a custom JSON-based protocol. I²C buses are available for ADC expansion modules and future sensor add-ons, while GPIO pins provide interrupt-driven input from user switches and lab safety sensors.

7. Audio and Vision Interface

7.1. Overview

The Audio and Vision Interface defines the data, signal, and software connections between the Jetson Orin Nano processing unit, the microphone/speaker array, and the camera subsystem. These interfaces enable real-time speech recognition (STT), text-to-speech (TTS) synthesis, multilingual translation, image-to-text processing, and multimodal fusion for question-answering.

7.2. Speech Interface

7.2.1. Input Hardware

Dual digital MEMS microphones connected via USB audio interface to the Jetson Orin Nano.

7.2.2. Output Hardware

4 Ω / 3 W speaker-amplifier module powered from the 5 V rail of the Power PCB.

7.2.3. **Software Stack**

OpenAI Whisper STT engine → Local LLM + RAG backend → TTS synthesis (via Coqui TTS or PyTorch Vocoder).

7.2.4. Signal Path

Signal Path:

- Analog audio → ADC (USB audio codec) → Jetson audio driver → Whisper STT.
- AI response → TTS engine → DAC → amplifier → speaker output.

7.2.5. Data Format

16-bit PCM, 44.1 kHz sampling rate.

7.2.6. **Latency Requirement**

End-to-end (STT→LLM→TTS) response ≤ 1.0 s for typical single-sentence queries.

7.3. Censorship and Translation Layer

7.3.1. Censorship

Real-time lexical filter applied post-STT; profanity or restricted terms are masked before forwarding to the LLM.

7.3.2. Translation

Integrated multilingual model detects language from STT output and performs automatic translation to English (or maintains native language for echo playback).

7.3.3. **Interface Dependency**

Operates entirely in software on the Jetson CPU / GPU; no additional hardware interface required.

7.3.4. Database Logging

Filtered and translated transcripts stored in the local SQLite database with timestamps.

7.4. Vision Interface

7.4.1. Input Hardware

8 MP CSI or USB camera connected to Jetson Orin Nano.

7.4.2. Processing Stack

Segment Anything Model (SAM 2) + OpenCV + local LLM for image captioning and contextual query interpretation.

7.4.3. Multimodal Fusion

STT text and image embedding combined through LLM token alignment to support speech + image questions.

7.4.4. Output

Textual or spoken AI response rendered via TTS module and UI display.

7.2.5. Frame Format

YUV or RGB 640 × 480 minimum, 30 FPS.

7.5. Communication Interfaces

7.5.1. Jetson ↔ ESP32

UART (JSON commands) for voice-triggered robotic actions

7.5.2. Jetson ↔ Database

Local SQLite or MySQL API for speech and image logs.

7.5.3. Jetson ↔ Tablet/Web UI

WebSocket or REST API for live speech display, translation mode toggle, and audio control.

7.6. Environmental and Electrical Parameters

Parameter	Value / Range	Notes
Microphone Operating Voltage	5 V DC	Draw < 50 mA per channel
Speaker Amplifier Voltage	5 V DC	Peak 1 A draw under speech output
Camera Input Voltage	5 V DC	< 200 mA draw
Audio Signal Type	Digital PCM USB	Isolated from main power noise
Operating Temperature	0–45 °C	Lab ambient conditions

Table 11: Environmental and Electrical Parameters

8. Artificial Intelligence Model

8.1. Overview

The Artificial Intelligence subsystem provides the core natural-language processing capabilities for AURA. It operates entirely on the Jetson Orin Nano and consists of a locally hosted large language model (LLM), an Ollama runtime environment, a Chroma vector database, and a custom LightRAG retrieval pipeline. The subsystem processes text-based user queries from the UI, retrieves relevant document context, and generates grounded responses.

The AI subsystem interfaces with the Website/UI subsystem through a local REST/WebSocket API and communicates with internal software modules for document updates, session handling, and logging. No external cloud dependencies are required.

8.2. Functional Inputs

The AI subsystem accepts the following inputs:

Input Type	Source	Format	Description
User Query Text	Website/UI	UTF-8 JSON string	Main text input from user for LLM processing
Document Uploads	Website/Admin	PDF or text file	Course materials used to build vector database
Session Control Commands	Website/Admin	JSON commands	Save, load, clear, or terminate sessions
Retrieval Parameters	Internal Config	JSON	Chunk size, top-k, LightRAG graph settings

Table 12: Acceptable AI Subsystem Inputs

8.3. Functional Outputs

Output Type	Destination	Format	Description
Text Response	Website/UI	UTF-8 JSON	Final LLM-generated answer to user query
Retrieved Document Chunks	Internal Logging	JSON array	Chunks selected by LightRAG for grounding
Session Files	Local Storage	JSON	Persistent conversation logs

Analytics Data	Developer Tools	JSON	Accuracy, response time, retrieval success

Table 13: Acceptable AI Subsystem Outputs

The AI subsystem outputs only text data and has no direct audio, vision, or motor interfaces.

8.4. Internal Data Flow

The internal data flow of the AI subsystem follows four sequential stages:

1. **Query Intake**
   User text is received via local API endpoints and formatted as a JSON request.
2. **Vector Retrieval (LightRAG)**
   The query is embedded, compared with the Chroma vector database, and re-ranked using LightRAG's graph-based relevance algorithm.
3. **Grounded LLM Generation**
   Retrieved context is injected into the LLM prompt and processed through the Ollama runtime.
4. **Response Packaging**
   The output is returned to the UI subsystem along with metadata such as retrieval statistics and session state.

This process is fully executed on the Jetson and requires no external computation.

8.5. Communication Interfaces

8.5.1 Jetson ↔ Website/UI Subsystem

Protocol: REST or WebSocket
Data Type: JSON
Function:

- Send user queries to AI subsystem
- Receive model responses
- Upload reference documents
- Request session operations

8.5.2 Jetson ↔ Local Storage

Protocol: File I/O
Format: JSON
Function:

- Save/load session history
- Store vector database files
- Log model performance and analytics

8.5.3 Jetson ↔ AI Runtime (Ollama)

Protocol: Local API (Ollama HTTP API / CLI)
Function:

- Load and run LLM models (e.g., Llama3.2 1B)
- Retrieve model capabilities and status

8.5.4 Jetson ↔ Vector Database (ChromaDB)

Protocol: Local embedded database API
Function:

- Store embeddings
- Retrieve top-k vector matches
- Manage document collections

8.6. Data Formats and Parameters

Parameter	Value / Format	Notes
Embedding Dimensionality	Model-dependent	Used for vector indexing
Chunk Size	512 tokens	Adjustable via config
Top-k Retrieval	6	Default for LightRAG

Session File Format	JSON object	Stores conversation history
Model API Input	Text prompt	UTF-8 encoded
Model API Output	Text response	Includes metadata

Table 14: Data Formats and Parameters

8.7. Performance and Timing Requirements

Requirement	Target Value	Notes
Time to First Token	≤ 5 seconds (Jetson target)	CPU version slower; GPU improves significantly
Retrieval Success Rate	≥ 95%	Achieved 100% in validation
Correctness Accuracy	≥ 85%	Achieved 90% on ECEN 214 dataset
Session Load Time	≤ 0.5 seconds	JSON-based persistence

Table 15: Performance and Timing Requirements

8.8. Performance and Timing Requirements

- Must operate fully offline.
- Must only produce responses grounded in uploaded documents.
- Must maintain deterministic behavior across identical inputs.
- Must log all interactions for review.
- Must not control motors or actuators directly.

8.9. Interface Dependencies

Dependent Subsystem	Dependency	Description
Website/UI	API requests	Sends queries, receives responses
Power System	Stable 10.5 V rail (Jetson)	Ensures continuous inference
Storage	Session + DB files	Required for persistence
Admin Tools	Document management	Updating or replacing corpus materials

Table 16: Interface Dependencies